Data stream

Daily national consumption (REN)

2015-2025, 3,926 days

Weather of 6 Portuguese cities

(Open-Meteo), calendar, lags

Horizons: 1 and 7 days

Predict → score → update

AR-ENFS

Evolving Takagi-Sugeno rule base

Conjunctive Gaussian activation

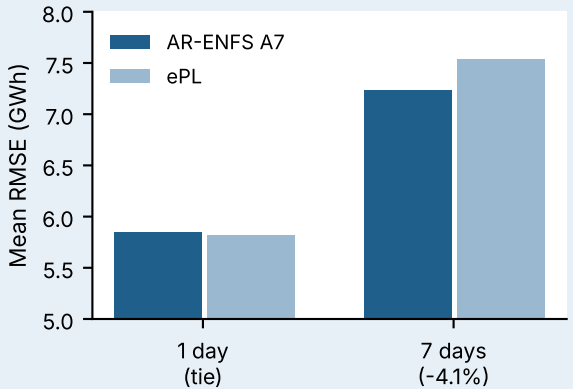
Direct + residual consequents (WRLS)

Seasonal regimes on consequents

Novelty AND (error OR uncertainty)

→ bounded rule growth

Findings



Rules = warm / cool-season regimes

Trade-off: more sensitive to input noise